

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	6ab7f10630f9767f87728ca3
Project Name	Import Data Using Transform Maps (Spreadsheet)
Team Leader	E Dilli Kumar
Team Members	R Vinish, K Jerina, B Mohanavalli, P Sri Ragavan
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how a spreadsheet is imported into the target table using an Import Set and a Transform Map. The user uploads the file, the data lands in a staging (import set) table, the Transform Map maps and cleans each row into the target table, and results and errors are logged for review.

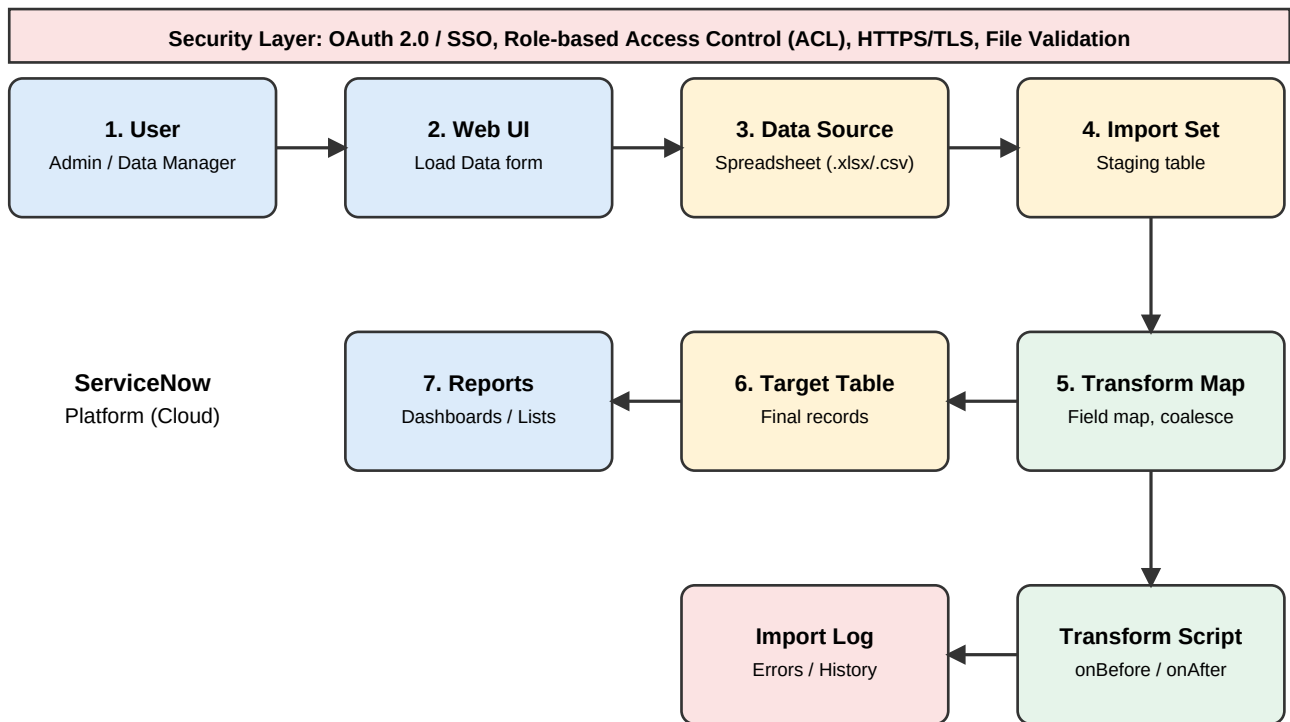


Figure 1: Architecture of Import Data Using Transform Maps (Spreadsheet)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface where the user selects the spreadsheet, uploads it (Load Data) and views the import and transform results.	HTML, CSS, JavaScript; ServiceNow Next Experience UI / Service Portal
2	Application Logic-1	Reads the spreadsheet, creates the import set (staging) table and loads the rows.	ServiceNow Import Sets, Data Source (Excel / CSV)
3	Application Logic-2	Transform Map: maps source columns to target fields, coalesces on key fields, inserts or updates records.	ServiceNow Transform Map, Field Maps, Coalesce

S.No	Component	Description	Technology
4	Application Logic-3	Custom data conversion and validation (data cleaning, reference lookups, onBefore / onAfter logic).	JavaScript (Transform Scripts, Field Map Scripts)
5	Database	Stores the staging (import set) table and the target table records.	ServiceNow relational database (MariaDB / MySQL based)
6	Cloud Database	Cloud-hosted instance database that holds all imported data and import logs.	ServiceNow Cloud Instance Database
7	File Storage	Temporary storage of the uploaded spreadsheet file (attachment) before processing.	ServiceNow Attachments (sys_attachment) / Local filesystem
8	External API-1	Optional: import of data from external sources and systems through REST for scheduled imports.	ServiceNow REST API / Scheduled Import
9	External API-2	Optional: pre-processing of the spreadsheet (xlsx to csv conversion) and notification through email.	Apache POI / Python pandas (pre-processing), SMTP Email
10	Machine Learning Model	Not used in the current version. Future scope: predictive field mapping and duplicate detection.	N/A (Future: ServiceNow Predictive Intelligence)
11	Infrastructure (Server / Cloud)	Application deployed on the cloud; developed and tested on a Personal Developer Instance (PDI). Local Server Configuration: Windows / Linux laptop with a web browser. Cloud Server Configuration: ServiceNow cloud instance.	ServiceNow PDI, Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source libraries used for the front end, file handling and pre-processing of spreadsheet data.	JavaScript, Bootstrap, AngularJS (Service Portal), Python pandas, Apache POI
2	Security Implementations	Role-based access control for who can import and transform data, authenticated login, encrypted transfer and validation of the uploaded file type and size.	ACLs / IAM roles, OAuth 2.0, SSO, HTTPS/TLS, OWASP guidelines, SHA-256
3	Scalable Architecture	Multi-tier architecture (presentation, application, database). Large files are processed in batches and new transform maps and tables can be added without redesign.	3-tier architecture, Scheduled / Bulk Import, Transform Maps
4	Availability	Cloud-hosted instance with redundant data centres, load balancing, regular backups and failover.	ServiceNow Cloud (load balancers, data-centre replication, backups)
5	Performance	Optimised for fast imports through batch processing, indexed coalesce fields, asynchronous transforms and caching of lookups.	Asynchronous Transform, Indexing, Cache, CDN

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

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<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>